

Integrability of Subsampled Compositional Diffusions

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Abstract

Compositional diffusion inference combines many factor scores, making subsampling attractive when evaluating every factor is expensive. For a Feynman–Kac correction, however, an unbiased estimate of the instantaneous potential is exponentiated to obtain a positive particle weight. We study whether the resulting population operator has a finite normalizing constant. For Gaussian factors, we give an exact reachable-variance recursion and reduce its validity check to two extreme batches per time step, independently of every finite batch size. A four-factor example has a finite full-factor update and an infinite subsampled normalizer for every finite batch size, even as the local estimator mean squared error tends to zero. We then construct an unbiased control-variate estimator that preserves the deterministic quadratic tail of equal-component-variance Gaussian mixtures. Its strict integrability domain agrees with that of the full-factor discretization. Controlled oracle-score experiments examine the separation between population integrability, finite-particle behavior, and approximation error. The results identify tail curvature as a validity condition for exponential subsampling, alongside the usual variance and computational-cost criteria.

1 Introduction

Compositional inference uses separately available conditional distributions to build a posterior for multiple observations or groups. This structure is useful in simulation-based inference, where amortized factor models can be reused across datasets. Recent work develops diffusion posterior sampling for tall data (Linhart et al., 2026), and compositional amortized inference for large hierarchical models (Arruda et al., 2026). Evaluating all factor scores at each particle and diffusion time can nevertheless dominate inference cost. A natural computational question is whether a small random batch of factor scores can replace the full sum.

That question becomes delicate when composition is accompanied by Feynman–Kac weighting. The correction contains a quadratic expression in the composed score. An unbiased estimator of this expression is straightforward to construct with independent, uniformly sampled factor indices. Its exponential is positive, and any finite collection of sampled states produces finite real-valued log weights in exact arithmetic. Neither property guarantees that the population weight can be integrated. A rare batch can create an excessively positive quadratic potential in a Gaussian tail, making its expected exponential infinite.

This paper studies that existence question for a specified weighted Euler discretization. The issue is separate from the established mismatch between composed intermediate distributions and forward-noised target distributions, and from learned score error (Linhart et al., 2026; Soiffer et al., 2026). We use exact factor scores so that these additional approximation sources do not obscure the effect of random exponential weights. We also distinguish finite normalizing constants from small endpoint error: a valid discretization may still be inaccurate, and a finite-particle sample may appear accurate when its population operator is undefined.

Our main result makes this distinction computable. For scalar Gaussian factors with non-negative residual precisions, the largest variance reachable over all finite batch histories evolves by a two-choice recursion. The choices are batches containing only the weakest or

only the strongest factor. These events have positive probability for every finite batch size, so increasing the batch size can reduce estimator mean squared error without changing the integrability certificate. A closed-form four-factor example exhibits this separation in a single step. For non-Gaussian factors with a shared variance within each Gaussian mixture, we preserve the deterministic quadratic tail through an affine control variate. We prove equality of the strict integrability domains of this estimator and the full-factor update. The empirical study reports both the certificates and the observed particle errors.

2 Compositional flow and random exponential weights

Let φ be the standard Gaussian density on $\mathbb{R}^d$. Factor density $q_{g,0}$ evolves under the variance-preserving Ornstein–Uhlenbeck process

$$dX_u = -\frac{1}{2}X_u du + dW_u, \quad X_u = e^{-u/2}X_0 + \sqrt{1 - e^{-u}}\xi. \quad (1)$$

Write $s_g(x, u) = \nabla \log q_{g,u}(x)$, $r_g = s_g + x$, and $R = \sum_{g=1}^G r_g$. The unnormalized compositional path is

$$\rho_u(x) = \varphi(x)^{1-G} \prod_{g=1}^G q_{g,u}(x), \quad \nabla \log \rho_u = -x + R. \quad (2)$$

We assume this path is integrable over the interval considered. With reverse time $t = U - u$, define

$$b_\kappa(x, u) = \frac{1 + \kappa}{2}R(x, u) - \frac{\kappa}{2}x, \quad g(x, u) = \frac{1}{2} \left(\|R(x, u)\|^2 - \sum_{g=1}^G \|r_g(x, u)\|^2 \right), \quad (3)$$

where $\kappa > 0$. Direct differentiation gives

$$-\partial_u \rho_u = -\nabla \cdot (b_\kappa \rho_u) + \frac{\kappa}{2} \Delta \rho_u + g \rho_u. \quad (4)$$

Thus the normalized continuous flow is represented by a Feynman–Kac diffusion whenever its required moments exist. Appendix A derives this identity. Our results concern the following discrete operator; they do not assert that its finite-step target equals ρ_0 .

For a step $h > 0$ at level u , independently draw a batch $B = (I_1, \dots, I_M)$, with each index uniform on $\{1, \dots, G\}$. For $M \geq 2$, define

$$\widehat{R}_B = \frac{G}{M} \sum_{i=1}^M r_{I_i}, \quad (5)$$

$$\widehat{g}_{U,B} = \frac{1}{2} \left\{ \frac{G^2}{M(M-1)} \sum_{i \neq j} r_{I_i}^\top r_{I_j} - \frac{G}{M} \sum_i \|r_{I_i}\|^2 \right\}. \quad (6)$$

Both are conditionally unbiased at every fixed (x, u) . The plug-in estimator $\widehat{g}_N = \frac{1}{2}(\|\widehat{R}\|^2 - \frac{G}{M} \sum_i \|r_{I_i}\|^2)$ is also considered. Its conditional bias is $G^2 \text{tr Cov}_I(r_I)/(2M)$.

For either estimator, let $\widehat{b}_B = (1 + \kappa)\widehat{R}_B/2 - \kappa x/2$. The positive kernel and its normalized update are

$$Q_{u,h}f(x) = \mathbb{E}_{B,\xi} \left[e^{h\widehat{g}_B(x,u)} f\left(x + h\widehat{b}_B(x, u) + \sqrt{\kappa h}\xi\right) \right], \quad \Phi_{u,h}(\eta)(f) = \frac{\eta Q_{u,h}f}{\eta Q_{u,h}1}. \quad (7)$$

The denominator must belong to $(0, \infty)$. Conditional unbiasedness of $\widehat{g}$ does not imply $\mathbb{E}_B e^{h\widehat{g}} = e^{hg}$ or that $\eta Q1$ is finite. We analyze this denominator before making any particle-convergence claim.

3 A batch-size-independent Gaussian certificate

In one dimension, Gaussian factors have $r_g(x, u) = -a_g(u)x + d_g(u)$. At a fixed level, suppress u and suppose $a_g \geq 0$. For a batch, put $m = M^{-1} \sum_i a_{I_i}$ and $s^2 = M^{-1} \sum_i (a_{I_i} - m)^2$. The drift and potential have the form

$$\hat{b}_B(x) = -\frac{1}{2}((1 + \kappa)\hat{A}_B + \kappa)x + \text{constant}, \quad \hat{g}_B(x) = \hat{c}_B x^2 + \text{linear terms}, \quad (8)$$

where $\hat{A}_B = Gm$, and

$$\hat{c}_{U,B} = \frac{1}{2} \left[G(G-1)m^2 - \left(\frac{G^2}{M-1} + G \right) s^2 \right], \quad \hat{c}_{N,B} = \frac{1}{2} [G(G-1)m^2 - Gs^2]. \quad (9)$$

For the full-factor update, $A = \sum_g a_g$ and $c = (A^2 - \sum_g a_g^2)/2$.

Lemma 1 (Conditional Gaussian update). Suppose the normalized state conditional on a batch history is Gaussian with variance $V > 0$. Its next conditional normalizer is finite if and only if

$$D_B(V) = 1 - 2h\hat{c}_B V > 0. \quad (10)$$

When finite, the propagated conditional law is Gaussian with variance

$$F_B(V) = \frac{[1 - h((1 + \kappa)\hat{A}_B + \kappa)/2]^2 V}{D_B(V)} + \kappa h. \quad (11)$$

The criterion and variance are independent of the Gaussian mean and the linear terms in the potential.

Proof. Multiplication by the weight changes the Gaussian precision from V^{-1} to $V^{-1} - 2h\hat{c}_B$. A nonpositive precision gives an infinite integral on $\mathbb{R}$, including the zero-precision case. For positive precision, completing the square gives tilted variance $V/D_B(V)$. The affine Euler map and independent Gaussian noise give (11). $\square$

Every batch history has positive probability. If one of its conditional normalizers is infinite, Tonelli's theorem makes the joint nonnegative integral infinite. If all histories remain finite for a finite number of steps, their finite sum is finite. A direct check appears to require exponentially many histories. The following result removes that enumeration.

Theorem 2 (Two extreme batches suffice). Fix a finite time grid, $G \geq 2$, $M \geq 2$, and $\kappa > 0$. Assume scalar Gaussian factors with $a_g(u_k) \geq 0$ and an initial Gaussian variance $V_0 > 0$. For either estimator in (9), set $V_{\max,0} = V_0$. At step k , let $a_- = \min_g a_g(u_k)$, $a_+ = \max_g a_g(u_k)$ and define, for $z \in \{a_-, a_+\}$,

$$D_z = 1 - h_k G(G-1)z^2 V_{\max,k}, \quad F_z = \frac{[1 - h_k((1 + \kappa)Gz + \kappa)/2]^2 V_{\max,k}}{D_z} + \kappa h_k. \quad (12)$$

If some $D_z \leq 0$, the population normalizer is infinite at this step. Otherwise set $V_{\max,k+1} = \max(F_{a_-}, F_{a_+})$ and continue. Until failure, this is exactly the largest reachable conditional variance. Consequently the integrability decision is identical for every finite $M \geq 2$, for both the plug-in and unbiased estimators.

Proof. Both coefficients in (9) are at most $G(G-1)m^2/2$, with equality for a constant batch. Since $0 \leq m \leq a_+$, a nonpositive denominator can occur if and only if it occurs for the constant strongest-factor batch at the largest reachable variance. On the common positive-denominator domain, $F_B(V)$ is nondecreasing in V and in $\hat{c}_B$ with $\hat{A}_B$ fixed. It remains to maximize

$$f(m) = \frac{V[1 - h((1 + \kappa)Gm + \kappa)/2]^2}{1 - hVG(G-1)m^2}, \quad m \in [a_-, a_+].$$

This function is convex. Indeed, with affine $n(m) = \sqrt{V}[1 - h((1 + \kappa)Gm + \kappa)/2]$ and positive concave $d(m) = 1 - hVG(G-1)m^2$,

$$f(m) = \sup_{z \in \mathbb{R}} \{2zn(m) - z^2 d(m)\},$$

the supremum of convex quadratics. Its maximum on the interval occurs at an endpoint. The corresponding constant batches exist for every M and have positive probability. Induction establishes attainability of each maximum and the stated decision. $\square$

The cost is $O(KG)$ to obtain the extrema and $O(K)$ scalar transitions, with no enumeration over G^M batches. For diagonal product factors, apply the recursion to each coordinate: a single failing coordinate suffices for divergence, while positivity for every coordinate certifies every history.

3.1 An exact four-factor separation

Take zero-mean Gaussian factor precisions $1 + \lambda_g$ with $\lambda = (1/10, 1/5, 2, 4)$. At $u = \log 2$, the residual precisions are $a = (1/21, 1/11, 1/2, 2/3)$. The normalized compositional density is $\mathcal{N}(0, V)$ with $V = 154/355$. For $h = 1/2$ and $\kappa = 1$,

$$A = \frac{201}{154}, \quad c = \frac{346}{693}, \quad D_{\text{full}} = \frac{2503}{3195} > 0, \quad D_{\text{all strongest}} = -\frac{167}{1065} < 0. \quad (13)$$

The all-strongest batch occurs with probability 4^{-M} and has $\hat{c} = 8/3$ for either estimator. Thus the subsampled normalizer is infinite for every finite M , whereas the full-factor update has log normalizer $-\frac{1}{2} \log(2503/3195)$ and output variance

$$[1 - \frac{1}{2}(A + \frac{1}{2})]^2 \frac{154/355}{2503/3195} + \frac{1}{2}. \quad (14)$$

At each fixed state, both unbiased estimators converge in mean square as $M \rightarrow \infty$, because the index distribution has finite support. Their state-averaged mean squared errors also vanish for this Gaussian input. This limit does not interchange with the nonnegative exponential integral. For P independent particle batches, the probability of observing at least one all-strongest batch is $1 - (1 - 4^{-M})^P$; observing such a batch still does not guarantee an extreme sampled state.

4 Preserving the deterministic quadratic tail

Suppose each scalar factor is a finite Gaussian mixture whose components share variance v_g but may have different means. After forward noising, its residual score admits

$$r_g(x, u) = r_g^0(x, u) + e_g(x, u), \quad r_g^0 = -a_g(u)x + d_g(u), \quad \sup_x |e_g(x, u)| < \infty, \quad (15)$$

where $a_g(u) = [1 - e^{-u} + e^{-u}v_g]^{-1} - 1$. One may use $d_g = e^{-u/2}\bar{\mu}_g/[1 - e^{-u} + e^{-u}v_g]$, with $\bar{\mu}_g$ the mixture mean. The bound follows because the posterior component mean lies in the convex hull of finitely many component means. A Gaussian fit using the total mixture variance generally does not satisfy this bounded-residual property.

Write $R_0 = \sum_g r_g^0$ and $Q_0 = \sum_g \|r_g^0\|^2$. Compute both affine/quadratic functions exactly. The tail-control estimator is

$$\begin{aligned} \hat{R}_T &= R_0 + \frac{G}{M} \sum_i e_{I_i}, \\ \hat{g}_T &= \frac{1}{2}(\|R_0\|^2 - Q_0) + R_0^\top \frac{G}{M} \sum_i e_{I_i} - \frac{G}{M} \sum_i (r_{I_i}^0)^\top e_{I_i} \\ &\quad + \frac{1}{2} \left\{ \frac{G^2}{M(M-1)} \sum_{i \neq j} e_{I_i}^\top e_{I_j} - \frac{G}{M} \sum_i \|e_{I_i}\|^2 \right\}. \end{aligned} \quad (16)$$

Taking conditional expectations proves unbiasedness. Expanding $r_g = r_g^0 + e_g$ gives exactly the full potential. Moreover, every batch has the same quadratic coefficient c and the same linear drift coefficient as the full-factor construction; all random contributions to the potential grow at most linearly.

Theorem 3 (Strict integrability domain of tail control). Consider scalar equal-component-variance Gaussian mixture factors, a finite grid with $\kappa > 0$, and an initial density satisfying

$$\log p_0(x) = -x^2/(2V_0) + O(1 + |x|). \quad (17)$$

At each step propagate the full-factor quadratic variance using (11) with (A, c) . If all resulting denominators are strictly positive, both the full-factor and tail-control population normalizers are finite. If the first nonpositive denominator is strictly negative, both normalizers are infinite there. This statement holds for every finite $M \geq 2$. No claim is made at a zero denominator for general mixtures.

Proof. For a density with (17) and a potential $cx^2 + O(1 + |x|)$, weighting replaces its quadratic variance by $V/(1 - 2hcV)$ when the denominator is positive; a negative denominator forces an infinite integral. An affine map plus a bounded displacement followed by independent Gaussian noise preserves the quadratic-tail form, with variance (11); Appendix B proves this step. Equation (16) makes these quadratic coefficients common to every batch. Induction over the finite histories gives the result. $\square$

The theorem certifies existence of the discretized population flow. It does not remove Euler error or the effect of exponentiating the bounded residual estimator. Its computational requirement is also explicit: the affine coefficient sums can be prepared once per time level in $O(Gd)$ work, followed by M nonlinear factor evaluations per particle. The control requires known component tail variances, which limits its direct applicability to arbitrary learned score models.

More generally, an arbitrary affine control also admits an exact strict-domain certificate: it suffices to evaluate the G constant batches, each containing one repeated factor. Appendix C proves this extension and shows that the paired correction has the same strict integrability domain. Thus the moment-matched and tail-matched controls in our experiments can be classified using the same test, although only the latter preserves the full-factor domain by construction.

5 Experiments

The fixed protocol comprises 1,260 main cells and 720 supplementary cells. It uses exact Gaussian and Gaussian-mixture factor scores, isolating random-weight effects from learned-model approximation. All benchmark cells use 8,192 particles, $M = 4$, $G \in \{16, 64\}$, $d \in \{1, 8\}$, $K \in \{128, 512, 2048\}$, and five random seeds. The reverse-time grid is $u_k = 20(1 - k/K)^2$, the initial distribution is $\mathcal{N}(0, I)$, and $\kappa = 1$. The finite- U initialization is an approximation to the exact compositional path. Reference marginals are computed by numerical quadrature on 65,537 points in $[-12, 12]$.

The main comparison includes full-factor weighting, an unweighted composed drift, plug-in weighting, unbiased quadratic weighting, a paired cumulant correction, and two moment-matched Gaussian control variants. The supplementary benchmark evaluates tail control and its paired correction on both mixture families. Paired corrections use two independent batches and the positive weight $\exp\{h(\hat{g}_1 + \hat{g}_2)/2 - h^2(\hat{g}_1 - \hat{g}_2)^2/8\}$, with the averaged drift; they therefore require twice as many sampled factor evaluations. This correction is an empirical ablation, with no claim of exact exponential unbiasedness.

For the one-step example in (13), the supplementary study uses $P \in \{1,024, 16,384, 262,144, 1,048,576\}$, $M \in \{2, 4, 8, 16, 32\}$, and 20 seeds, together with the full-factor update. We retain histograms and the 1,024 largest normalized particle weights for each one-step cell. The benchmark retains complete particle arrays and reference distributions.

All 1,980 cells completed on SCNet RTX 3080 GPUs. We verified the source versions, 13,506 recovered file hashes, all 40 task exit records, and the complete cell configuration lists. Benchmark metrics were recomputed using independent numerical routines. Adaptive quadrature over the real line also verified all 54 reference coordinates, including analytic

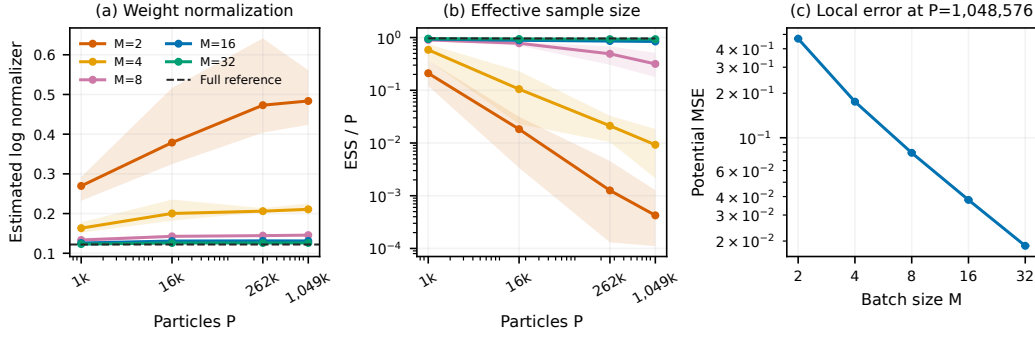


Figure 1: One-step Gaussian counterexample. Panels (a,b) show medians and interquartile ranges over 20 seeds. The dashed reference is the exact full-factor log normalizer in (a) and the empirical full-factor ESS fraction in (b). Panel (c) shows mean potential MSE with one standard deviation at $P = 1,048,576$. Every subsampled curve has an infinite population normalizer.

Gaussian checks. These are controlled oracle experiments; no learned score network is trained or evaluated.

5.1 Local error decreases while the normalizer remains infinite

Figure 1 shows the one-step experiment. At $P = 1,048,576$, increasing M from 2 to 32 reduces mean potential MSE from 0.46798 to 0.01857, approximately a factor of 25. The median ESS fraction rises from 0.000424 to 0.932706. Nevertheless, (13) gives an infinite population normalizer for both batch sizes and every intermediate finite M .

For $M = 32$, the empirical mean log normalizer is 0.126047 with standard deviation 0.000259 across 20 seeds. The full-factor result is 0.122030 ± 0.000234 , close to its exact value 0.122049. No nonintegrable batch was sampled in these 20 runs of 1,048,576 particles with $M = 32$. The all-strongest event alone has probability $4^{-32} \approx 5.42 \times 10^{-20}$ per batch. Its absence in this finite experiment cannot establish existence of the population target. For $M = 2$, the median estimated log normalizer rises from 0.270 at $P = 1,024$ to 0.484 at $P = 1,048,576$, accompanied by severe ESS loss. The proof supplies the divergence result; these finite observations illustrate why it can be visible at small M and inconspicuous at larger M .

5.2 Certificates across the benchmark grid

Table 1 reports validity over the 12 configurations per family, formed by two group counts, two dimensions, and three time grids. Seeds do not change a certificate. Raw plug-in, unbiased, and paired estimators fail all Gaussian and ordinary-mixture configurations. Moment matching makes the Gaussian estimator exact, but does not certify any ordinary-mixture configuration. Tail matching recovers the full-factor strict domain in both mixture families. The complete configuration-level grid is provided in Appendix F.

The full-factor discretization also fails for $G = 64, K = 128$ in the Gaussian and ordinary-mixture families. Its Gaussian denominator first becomes negative at step 100, reaching approximately -0.003111 . Yet the observed mean marginal W1 is only 0.0141 in one dimension and 0.0176 in eight dimensions. Thus a small endpoint sample discrepancy can coexist with an undefined population update even before subsampling. For finite Gaussian configurations, exact weighted-Euler moment recursion separates time error from particle error: at $G = 64, K = 2048$, the population W1 is approximately 0.000493, while the eight-dimensional finite-particle mean marginal W1 is 0.0149.

Table 1: Number of configurations with finite population normalizers, out of 12 per family. A dash denotes an unrun method; tail matching coincides with the Gaussian control on Gaussian factors. Unweighted updates are Markov kernels with unit weights, so finiteness alone is immediate and does not establish target accuracy.

Method	Gaussian	Mixture	Weak mixture
Full	10/12	10/12	12/12
Unweighted	12/12	12/12	12/12
Plug-in	0/12	0/12	4/12
Unbiased	0/12	0/12	4/12
Unbiased + pair	0/12	0/12	4/12
Moment CV	10/12	0/12	12/12
Moment CV + pair	10/12	0/12	12/12
Tail CV	–	10/12	12/12
Tail CV + pair	–	10/12	12/12

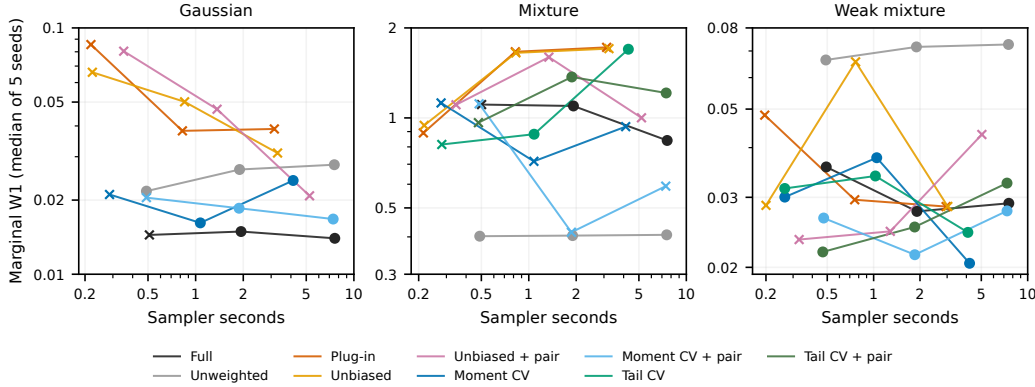


Figure 2: Observed error versus recorded sampler time at $G = 64, d = 8$, with $K \in \{128, 512, 2048\}$. Each point is a median over five seeds. Circles denote finite population normalizers and crosses denote infinite ones. Lines connect a method’s three time grids and are not fitted performance models.

5.3 A finite domain does not guarantee accurate or faster inference

Figure 2 and Table 2 compare actual sampler time and endpoint error. For the difficult eight-dimensional ordinary mixture with $G = 64, K = 512$, all weighted methods have substantial sampling error. Full-factor W1 is 0.881 ± 0.372 , tail control is 1.004 ± 0.327 , and paired tail control is 1.358 ± 0.188 . Their mean total-variation errors over the 256 coordinate-sign patterns are respectively 0.961, 0.975, and 0.998, confirming poor joint-mode coverage. The paired moment control has lower marginal W1, 0.456 ± 0.228 , despite an infinite population normalizer. These outcomes demonstrate that finite integrability, marginal accuracy, and joint coverage are distinct properties.

On the weak mixture, paired tail control yields W1 0.025 ± 0.005 , compared with 0.028 ± 0.007 for full weighting, at nearly the same recorded sampler time. Unpaired tail control uses 16 times fewer nonlinear particle-factor evaluations and about 1.86 times less sampler time, but its W1 rises to 0.043 ± 0.026 . We therefore make no general speed-accuracy superiority claim. Gaussian full and exact-control results, complete seed-level metrics, and additional joint diagnostics are retained in the artifacts.

Table 2: Matched configuration $G = 64, d = 8, K = 512$. W1 is mean $\pm$ one standard deviation over five seeds; seconds are medians. A superscript ∞ marks an infinite population normalizer. All methods use 8,192 particles.

Method	Mixture		Weak mixture	
	W1	Seconds	W1	Seconds
Full	0.881 ± 0.372	1.921	0.028 ± 0.007	1.909
Unweighted	0.404 ± 0.003	1.897	0.072 ± 0.001	1.896
Plug-in	$1.662 \pm 0.021^\infty$	0.822	$0.053 \pm 0.051^\infty$	0.754
Unbiased	$1.649 \pm 0.010^\infty$	0.828	$0.070 \pm 0.038^\infty$	0.764
Unbiased + pair	$1.608 \pm 0.099^\infty$	1.341	$0.032 \pm 0.014^\infty$	1.282
Moment CV	$0.742 \pm 0.097^\infty$	1.077	0.052 ± 0.040	1.047
Moment CV + pair	$0.456 \pm 0.228^\infty$	1.865	0.022 ± 0.005	1.850
Tail CV	1.004 ± 0.327	1.081	0.043 ± 0.026	1.027
Tail CV + pair	1.358 ± 0.188	1.876	0.025 ± 0.005	1.847

6 Related work

Compositional diffusion methods address scalable posterior inference and the behavior of products of score models (Linhart et al., 2026; Arruda et al., 2026; Soiffer et al., 2026). Our setup isolates an additional failure introduced by replacing a deterministic score-derived potential with an exponentiated random estimator. The compositional Feynman–Kac identity and the existence of path mismatch are background ingredients.

Random-weight particle filters have long treated positive stochastic weights explicitly (Fearnhead et al., 2010). Convergence with unbounded importance weights requires moment conditions (Mbalawata & Särkkä, 2016), and particle stability analyses impose assumptions on the underlying Feynman–Kac flow (Whiteley, 2013). Subsampling SMC uses approximately unbiased annealed likelihood estimators and auxiliary-variable updates (Gunawan et al., 2020). These works establish the relevance of positivity, moments, and exponentiation bias. The contribution here is the two-extreme-batch certificate for a particular compositional weighted Euler operator, its independence of every finite with-replacement batch size, and the preservation of its strict domain through deterministic tail curvature. The Gaussian integral and the unbiased quadratic estimator are classical calculations.

7 Discussion

The certificate distinguishes three questions that are often evaluated together: existence of a population target, accuracy of its time discretization, and behavior of a finite particle approximation. With-replacement subsampling can fail the first question while producing finite samples and decreasing local mean squared error. This sampling assumption is substantive: all without-replacement subset sizes in our four-factor one-step example have finite normalizers (Appendix E). Tail control preserves the full-factor strict integrability domain for the stated mixture class, but its accuracy and wall-clock cost remain empirical questions. The analysis is restricted to scalar or diagonal product tails and a fixed weighted Euler construction. Extending the certificate to noncommuting matrix covariances, adaptive batch selection, and learned models with certified tail behavior requires additional arguments.

AI use statement

An OpenAI coding agent was used for research ideation, literature searches, mathematical derivations, implementation, experiment orchestration, data analysis, and manuscript drafting. The automated verification accompanying this draft includes exact algebraic checks, numerical cross-checks, source and result provenance, and document compilation. This statement does not certify independent human review of the scientific claims. The authors must review the complete manuscript and artifacts before submission.

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A Derivation of the compositional Feynman–Kac identity

For a smooth positive factor density satisfying the Ornstein–Uhlenbeck forward equation,

$$\partial_u \log q_g = \frac{1}{2} \nabla \cdot r_g - \frac{1}{2} x^\top r_g + \frac{1}{2} \|r_g\|^2.$$

The prior φ is stationary. Consequently, with $Q = \sum_g \|r_g\|^2$,

$$-\partial_u \log \rho = -\frac{1}{2} \nabla \cdot R + \frac{1}{2} x^\top R - \frac{1}{2} Q.$$

Put $S = \nabla \log \rho = -x + R$. Since $b_\kappa = R/2 + \kappa S/2$,

$$\begin{aligned} \rho^{-1} [-\nabla \cdot (b_\kappa \rho) + \frac{\kappa}{2} \Delta \rho] &= -\nabla \cdot b_\kappa - b_\kappa^\top S + \frac{\kappa}{2} (\nabla \cdot S + \|S\|^2) \\ &= -\frac{1}{2} \nabla \cdot R + \frac{1}{2} x^\top R - \frac{1}{2} \|R\|^2. \end{aligned}$$

Adding $g = (\|R\|^2 - Q)/2$ yields (4). Normalization subtracts $\mathbb{E}_{\rho/\int \rho g}$ from the potential. This calculation identifies the continuous path; existence of the normalized stochastic representation requires the corresponding integrability assumptions.

B Gaussian tail envelopes under an Euler transition

We give the full scalar argument used by Theorem 3. Suppose

$$-\frac{x^2}{2V} - L|x| - C \leq \log p(x) \leq -\frac{x^2}{2V} + L|x| + C,$$

and $g(x) = cx^2 + r(x)$ with $|r(x)| \leq L_g(1 + |x|)$. Then

$$-\frac{D}{2V} x^2 - (L + hL_g)|x| - C' \leq \log[p(x)e^{hg(x)}] \leq -\frac{D}{2V} x^2 + (L + hL_g)|x| + C', \quad D = 1 - 2hcV.$$

For $D > 0$, the integral is positive and finite, and its normalization has the same quadratic envelope with $V_t = V/D$. For $D < 0$, the lower bound has a positive quadratic exponent and the integral diverges. At $D = 0$, the remaining terms can give either behavior, so the general statement excludes equality.

Now suppose the propagated state is

$$Y = AX + q(X) + \tau\xi, \quad |q(x)| \leq K, \quad \tau^2 = \kappa h > 0,$$

where X has the tilted density and $\xi \sim \mathcal{N}(0, 1)$ is independent. For $z = y - Ax$, the Gaussian transition-density ratio satisfies

$$e^{-K|z|/\tau^2 - K^2/(2\tau^2)} \leq \frac{\varphi_\tau(z - q(x))}{\varphi_\tau(z)} \leq e^{K|z|/\tau^2}.$$

Use the reference joint density $\varphi_{\sqrt{V_t}}(x)\varphi_\tau(y - Ax)$. Its Y marginal is Gaussian with variance $W = A^2V_t + \tau^2$, and its conditional distribution is

$$X | Y = y \sim \mathcal{N}\left(\frac{AV_t}{W}y, \frac{V_t\tau^2}{W}\right).$$

Multiplying the density-ratio bounds by the envelope for the tilted input reduces the output bounds to conditional expectations of $\exp\{\pm(L_1|X| + L_2|y - AX|)\}$. For fixed $L_1, L_2 \geq 0$, Gaussian exponential moments and the triangle inequality give an upper bound $e^{C_1|y| + C_2}$. Jensen's inequality gives the lower bound

$$\mathbb{E}[e^{-L_1|X| - L_2|y - AX|} | Y = y] \geq e^{-L_1\mathbb{E}[|X| | Y=y] - L_2\mathbb{E}[|y - AX| | Y=y]} \geq e^{-C_3|y| - C_4}.$$

It follows that $\log p_Y(y) = -y^2/(2W) + O(1 + |y|)$. The argument also covers $A = 0$ and bounded measurable q without a monotonicity assumption. On a fixed finite grid with a fixed finite batch size, the finitely many factors, batches, and histories admit common finite residual and envelope constants up to each strictly positive step. Every such conditional normalizer is finite and positive. Repeating the argument along these histories proves Theorem 3. For diagonal product models, a fixed index history leaves the coordinates conditionally independent, and the scalar argument applies coordinate by coordinate.

C A certificate for arbitrary affine controls

Suppose the true residual and its affine control have tail slopes $-a_g$ and $-t_g$, respectively. Set $e_g = a_g - t_g$, $T = \sum_g t_g$, and $c_0 = (T^2 - \sum_g t_g^2)/2$. The control-variate formula (16) remains conditionally unbiased when its residual is unbounded. For the empirical batch distribution p on the G factors, let $m = \sum_g p_g e_g$ and $s^2 = \sum_g p_g (e_g - m)^2$. Its quadratic coefficients are

$$\hat{A}(p) = T + Gm, \tag{18}$$

$$\hat{c}(p) = c_0 + GTm - G \sum_g p_g t_g e_g + \frac{1}{2}G(G-1)m^2 - \frac{1}{2}\left(\frac{G^2}{M-1} + G\right)s^2. \tag{19}$$

This identity follows by collecting the quadratic terms in (16); the bounded parts of mixture scores contribute only $O(1 + |x|)$ to the potential.

Proposition 4 (Constant-batch certificate). For Gaussian factors with arbitrary affine controls, the exact largest reachable variance is obtained by applying (11) to the G constant batches at each step and retaining the largest value. The corresponding integrability decision is independent of every finite $M \geq 2$. For equal-component-variance Gaussian mixture factors, the same recursion is exact for strict positive and strict negative denominator decisions under the tail-envelope assumptions of Theorem 3.

Proof. Delete the nonpositive variance term from (19), obtaining $\tilde{c}(p)$. This upper bound is convex on the probability simplex: its only quadratic term is the positive multiple $G(G-1)m^2/2$ of a squared affine function. Therefore $\tilde{c}(p) \leq \max_g \tilde{c}(\delta_g)$, and all vertex values are

attained by constant batches. If a denominator fails at the largest reachable variance, it fails at a vertex.

On the common positive-denominator domain, replacing $\hat{c}$ by $\tilde{c}$ increases the next variance. Write

$$n(p) = \sqrt{V}[1 - h((1 + \kappa)(T + Gm) + \kappa)/2], \quad d(p) = 1 - 2hV\tilde{c}(p).$$

Here n is affine and d is positive and concave. The quadratic perspective

$$n(p)^2/d(p) = \sup_{z \in \mathbb{R}} \{2zn(p) - z^2d(p)\}$$

is convex in p , so its maximum on the simplex is attained at a vertex. The upper bound is exact there because $s^2 = 0$. Monotonicity in V and induction complete the Gaussian proof. The mixture extension follows from Appendix B, since every fixed batch still has an affine drift plus a bounded term and a quadratic potential plus a linearly bounded term. $\square$

For the constant batch containing factor g , the coefficients used by the implementation are

$$A_g^* = T + Ge_g, \quad c_g^* = c_0 + G(T - t_g)e_g + \frac{1}{2}G(G - 1)e_g^2.$$

The check requires G candidates per step and has no combinatorial dependence on M . For exact tail slopes $t_g = a_g$, every $e_g = 0$, and all candidates coincide with the full-factor coefficients. With zero control, the simplex reduces to the scalar mean interval used in Theorem 2.

Corollary 5 (Paired correction preserves the strict certificate). Under the strict-domain conditions of Proposition 4, averaging two independent batch drifts and using the paired weight from Appendix D gives the same strict integrability decisions as the unpaired estimator. The statement also applies to the raw unbiased estimator.

Proof. The negative squared-difference correction can only reduce the weight. Omitting it gives quadratic potential coefficient $\bar{c} = (c_1 + c_2)/2$ and drift coefficient $\bar{A} = (A_1 + A_2)/2$. The variance expression

$$(A, c) \mapsto \frac{V[1 - h((1 + \kappa)A + \kappa)/2]^2}{1 - 2hVc} + \kappa h$$

is jointly convex in (A, c) on its positive-denominator domain, by the same quadratic-perspective identity. It is therefore at most the larger of the two unpaired next variances. Gaussian upper envelopes propagate under the omitted-correction transition; the actual positive weight obeys that upper bound after division by a finite positive normalizer. This proves finiteness along every paired history when the unpaired certificate stays strictly positive.

Conversely, take the maximizing constant batch at each step and use that same batch for both copies. This paired history has positive probability, the correction vanishes identically, and its tail variance follows the unpaired maximizing recursion. A strictly negative denominator therefore gives an infinite conditional integral. Tonelli’s theorem gives divergence of the full paired operator. Gaussian factors also allow the exact equality decision; the mixture result leaves equality unclassified. $\square$

D Second-order terms and paired weighting

This section records a local expansion to clarify the paired-correction ablation. Assume sufficient smoothness and integrable domination to justify expansion and normalization through third order for h in a neighborhood of zero. A bounded-state setup with bounded smooth coefficients is one sufficient setting. These assumptions must be checked before applying the expansion; it cannot justify a kernel with an infinite normalizer.

At a fixed time, suppose $\mathbb{E}_B \hat{b} = b$ and $\mathbb{E}_B \hat{g} = g$. Define $C_b(x) = \text{Cov}_B(\hat{b}(x))$, $c_{bg}(x) = \text{Cov}_B(\hat{b}(x), \hat{g}(x))$ and $v_g(x) = \text{Var}_B(\hat{g}(x))$. Let Q denote the random kernel and Q^0 the full-coefficient Euler kernel. Expansion of the unnormalized kernel yields

$$(Q - Q^0)f = h^2 \left\{ \frac{1}{2}C_b : \nabla^2 f + c_{bg}^\top \nabla f + \frac{1}{2}v_g f \right\} + O(h^3).$$

The diffusion-dependent terms are common to both kernels. Expanding the ratio in (7), the normalized population difference is

$$\Phi(\eta)(f) - \Phi^0(\eta)(f) = h^2 \left\{ \frac{1}{2} \eta(C_b : \nabla^2 f) + \eta(c_{bg}^\top \nabla f) + \frac{1}{2} \text{Cov}_\eta(f, v_g) \right\} + O(h^3). \quad (20)$$

The last term uses the covariance over the input population. A pointwise normalized kernel would be a different object. In particular, conditional score error alone does not specify this bias: its coupling with the potential enters through c_{bg} .

For two independent copies, use $\bar{b} = (\hat{b}_1 + \hat{b}_2)/2$ and weight

$$\exp \left\{ \frac{h}{2} (\hat{g}_1 + \hat{g}_2) - \frac{h^2}{8} (\hat{g}_1 - \hat{g}_2)^2 \right\}.$$

The expected squared difference is $2v_g$, cancelling the second-order potential-variance term. The remaining coefficient is

$$\frac{1}{4} \eta(C_b : \nabla^2 f) + \frac{1}{2} \eta(c_{bg}^\top \nabla f).$$

Thus paired weighting does not remove all second-order effects. For Gaussian factors, the event that both batches equal the same extreme batch at every step has positive probability; the correction vanishes on that history. Whenever Theorem 2 gives a divergent extreme history, this paired version also has an infinite normalizer.

E Without-replacement comparison

The with-replacement assumption is substantive. For a uniformly sampled subset of size $2 \leq M \leq G$, an unbiased potential estimator is

$$\hat{g}_{\text{WOR}} = \frac{G(G-1)}{M(M-1)} \sum_{i < j} r_{I_i}^\top r_{I_j}.$$

Distinct indices are sampled, so its pair coefficient differs from (6). For the same one-step example (13), exact subset enumeration gives the following maximum quadratic coefficients and minimum denominators:

M	2	3	4
$\max \hat{c}_{\text{WOR}}$	2	29/33	346/693
$\min D$	47/355	659/1065	2503/3195

All three without-replacement one-step operators are finite. At $M = G$ the estimator equals the full-factor potential. This analytic comparison shows that the failure established for every finite with-replacement M does not transfer to an arbitrary subsampling scheme. It does not assert multi-step stability of without-replacement updates for other grids or factor sets.

F Complete benchmark certificate

Every executed benchmark configuration is classified in Figure 3. The coordinate-wise tests classify the population operator before inspecting any particle outcome.

G Factor families and implementation details

For indices $g = 0, \dots, G-1$ and $j = 0, \dots, d-1$, set $\theta_{gj} = 2\pi(g + 0.37j)/G$. Each factor is a product over coordinates. The Gaussian family uses variance $v_{gj} = 0.5 + 0.15 \sin \theta_{gj}$ and mean $0.35 + 0.3 \cos \theta_{gj}$. The two-component mixture family uses the same component variance, component means

$$0.1 \cos \theta_{gj} \pm (0.7 + 0.2 \cos(2\theta_{gj})),$$

and positive-component weight $0.5 + 0.1 \sin(\theta_{gj} + 0.7)$. The weak mixture family uses

$$v_{gj} = \left[1 + \frac{4 + 2 \sin \theta_{gj}}{G} \right]^{-1}, \quad \mu_{gj,\pm} = \frac{0.4 \cos \theta_{gj}}{G} \pm \frac{0.6 + 0.1 \cos \theta_{gj}}{\sqrt{G}},$$

Population certificate: F = finite, I = infinite, - = not run

Gaussian, G=16, K=128	F	F	I	I	I	F	F	-	-
Gaussian, G=16, K=512	F	F	I	I	I	F	F	-	-
Gaussian, G=16, K=2048	F	F	I	I	I	F	F	-	-
Gaussian, G=64, K=128	I	F	I	I	I	I	I	-	-
Gaussian, G=64, K=512	F	F	I	I	I	F	F	-	-
Gaussian, G=64, K=2048	F	F	I	I	I	F	F	-	-
Mixture, G=16, K=128	F	F	I	I	I	I	I	F	F
Mixture, G=16, K=512	F	F	I	I	I	I	I	F	F
Mixture, G=16, K=2048	F	F	I	I	I	I	I	F	F
Mixture, G=64, K=128	I	F	I	I	I	I	I	I	I
Mixture, G=64, K=512	F	F	I	I	I	I	I	F	F
Mixture, G=64, K=2048	F	F	I	I	I	I	I	F	F
Weak mixture, G=16, K=128	F	F	I	I	I	F	F	F	F
Weak mixture, G=16, K=512	F	F	F	F	F	F	F	F	F
Weak mixture, G=16, K=2048	F	F	F	F	F	F	F	F	F
Weak mixture, G=64, K=128	F	F	I	I	I	F	F	F	F
Weak mixture, G=64, K=512	F	F	I	I	I	F	F	F	F
Weak mixture, G=64, K=2048	F	F	I	I	I	F	F	F	F

Full Unweighted Plug-in Unbiased Unbiased + pair Moment CV Moment CV + pair Tail CV Tail CV + pair

Figure 3: Complete certificate for the executed benchmark methods. The statuses agree for $d = 1$ and $d = 8$, so the dimensions share a row. Each row therefore represents ten executed cells per method across both dimensions and five seeds. The classification is derived from the population recursion, independently of particle outcomes.

with the same component weights. All component variances are below one, so their residual tail precisions are nonnegative. The Gaussian and ordinary mixture have identical tail certificates despite their different interior geometry.

At each step, weights are multiplied using the left-endpoint potential and states receive the Euler drift and Gaussian noise. Systematic resampling is performed when the effective sample size is below half the particle count. Computation uses double precision, one GPU per task, four allocated CPU cores, and one numerical-library thread. Timing includes sampling, weighting, and resampling, with GPU synchronization at the measured boundaries. Factor reference quadrature and final metric computation are outside the sampler timing. The score-evaluation count records particle-factor evaluations and counts both batches in paired methods.

The raw endpoint sample for each benchmark cell contains all particles and normalized weights. Reference grids, densities, cumulative distribution functions, step metrics, runtime metadata, configuration, and completion records accompany each cell. The one-step study retains weighted histograms and the largest-weight particles, as specified before execution. Independent verification recomputes benchmark metrics using separate SciPy routines and checks all recovered file hashes against the remote manifest.